

Lab 06 — Elastic Compute, Containers & Serverless

Build and run a containerised service with explicit CPU and memory limits, invoke a function-style endpoint, and inspect a batch worker.

Lab environment: Docker Desktop (free local simulation) — runs locally, no AWS account needed

Estimated time: 35–40 minutes

SAA-C03 domain: Design High-Performing Architectures

Overview

Local component	AWS equivalent
lab6-app (Flask, 0.5 CPU / 128 MB, port 8086)	An EC2 instance size or ECS task size
GET /function	An AWS Lambda function invocation
lab6-worker (loop every 10 s)	AWS Batch / an ECS scheduled task

Step 1 — Get the lab source and start the lab

The repo lives in `~/Downloads/aws-saa`. The paths are absolute, so these commands work from any folder. `git clone` is needed only once: if the repo is already there it prints `already exists`, and the next line still takes you into this lab.

```
cd ~/Downloads
git clone https://github.com/mohanpothula/aws-saa.git
cd ~/Downloads/aws-saa/labs/lab-06-compute-containers-serverless
docker compose up -d
docker compose ps
```

Step 2 — Build the image

Task: Build the Flask image.

```
docker compose build
```

Expected result: The build succeeds.

Step 3 — Run the service

Task: Start and test the app.

```
docker compose up -d
curl http://localhost:8086
```

Expected result: LAB6 Flask App is returned.

Step 4 — Inspect sizing

Task: Check the CPU and memory limits.

```
docker inspect lab6-app --format 'Memory={{.HostConfig.Memory}} NanoCPUs={{.HostConfig.NanoCpus}}'
```

Expected result: Memory=134217728 NanoCPUs=500000000 — 128 MB and half a CPU.

Step 5 — Function simulation

Task: Invoke the event endpoint.

```
curl http://localhost:8086/function
```

Expected result: `{"status": "processed"}` is returned.

Step 6 — Batch workload

Task: Inspect the worker logs.

```
docker compose logs --tail=5 worker
```

Expected result: `batch-job-<timestamp>` messages are visible.

Verification

Check	Expected
Step 2 — Build the image	The build succeeds
Step 3 — Run the service	LAB6 Flask App is returned
Step 4 — Inspect sizing	Memory=134217728 NanoCPUs=500000000 — 128 MB and half a CPU
Step 5 — Function simulation	<code>{"status": "processed"}</code> is returned
Step 6 — Batch workload	<code>batch-job-<timestamp></code> messages are visible

Completion check

- [] Every validation result demonstrated to the trainer
- [] You can say which AWS service each local component represents
- [] You can explain when you would pick Lambda over a long-running container

Cleanup

```
docker compose down -v
```

-v also removes the lab's volumes, so the next run starts from a clean state.

Reference

- Lab source code:
<https://github.com/mohanpothula/aws-saa/tree/main/labs/lab-06-compute-containers-serverless>
- Original student guide: [courseware/AWS_SAA_Lab6_Student_Guide.docx](#)
- [Amazon ECS Developer Guide](#)
- [AWS Lambda Developer Guide](#)
- [AWS Batch User Guide](#)
- [Docker Compose](#)
- [Compose file reference — deploy / resource limits](#)